

LEDGER · 2026-08-31 · ADVANCED MATERIALS

Shear-Exfoliated Few-Layer Graphene (FLG) via High-Speed Hydrodynamics

REJECTED

This concept did **not** clear the framework.
The reasoning is published in full below.
What the assessment found

Incumbent benchmark	Spherical Alumina
Entry application	High-performance Thermal Interface Materials (gap fillers) for EV battery packs
Measured advantage	~1.0–3.3 W/mK vs 2.0–4.0 W/mK for Spherical Alumina (negative delta <1x)
Time to first revenue	24 months
Gross margin at parity	-1875.0%
Startup CapEx	\$550,000
Capital productivity	0.08x
Regulatory risk	Low — no material regulatory CapEx reported (\$0 estimated)
Why it failed	

- Voided by: Capital and time to revenue, Economics undisclosed, Negative control / sub-2x, Entry application mismatch, Form-factor / handling regression

Assessment

The venture explicitly concedes failure due to negative gross margins and fundamental physical incompatibilities. Processing energy for solvent recovery pushes operating costs far above the \$6.00/kg incumbent price, while electrical conductivity creates catastrophic short-circuit hazards in EV battery packs.

Also flagged

- Price parity asserted, not sourced
- unsourced_equipment
- no_opex

A rejection is not a claim that the science is wrong. It means the venture case did not survive: the comparator was not what the buyer actually buys, the comparison was not like-for-like, the advantage fell short of a step change, or the capital and time to first revenue were prohibitive.

Assessed 2026-08-31 against seven gates plus the voiding sub-tests. [How the method works](#) · [Browse all concepts](#)

The Absence Audit

Method

Ledger

Free studies

Track record

Terms

Published for research and educational purposes. Nothing here is investment, legal, regulatory or engineering advice, and no commercial outcome is promised or implied. Concepts are drawn from published scientific literature and have not been commercially validated; several involve chemical, biological or regulated processes that require appropriate expertise, facilities, permits and safety controls. Independently verify every figure, citation and regulatory requirement before committing capital.

